Vezlo AI Assistant - Chatbot Flow Documentation

Detailed Flow Diagram

```
┌──────────────────────────────────────────────────────────────────────────┐
│                         USER QUERY ARRIVES                               │
│                    POST /api/messages/:uuid/generate                     │
└──────────────────────────────────────────────────────────────────────────┘
                                 │
                                 │
                                 ▼
┌──────────────────────────────────────────────────────────────────────────┐
│                      LOAD CONVERSATION CONTEXT                           │
│  • Fetch conversation by UUID                                            │
│  • Retrieve last N messages (CHAT_HISTORY_LENGTH=2)                     │
│  • Extract company_id (integer) for filtering                            │
└──────────────────────────────────────────────────────────────────────────┘
                                 │
                                 │
                                 ▼
┌──────────────────────────────────────────────────────────────────────────┐
│                  🧭 INTENT CLASSIFICATION (OpenAI)                        │
│  Model: AI_MODEL (gpt-4o-mini)                                           │
│  Input: User query + conversation history                                │
│  Output: Intent + Reason + Needs Guardrail + Contact Email              │
└──────────────────────────────────────────────────────────────────────────┘
                                 │
                                 │
                ┌────────────────┴─────────────────┐
                │                                  │
                │                                  │
                ▼                                  ▼
    ┌──────────────────────┐        ┌──────────────────────────┐
    │  SIMPLE INTENTS      │        │  KNOWLEDGE INTENT        │
    │  (Short-circuit)     │        │  (RAG Pipeline)          │
    └──────────────────────┘        └──────────────────────────┘
                │                                  │
    ┌───────────┴────────────┐                    │
    │                        │                    │
    │                        │                    │
    ▼                        ▼                    ▼
┌─────────┐         ┌──────────────┐    ┌────────────────────┐
│greeting │         │ personality  │    │ KNOWLEDGE LOOKUP   │
│         │         │              │    └────────────────────┘
│Response:│         │Response:     │              │
│"Hello!" │         │"I'm [NAME]"  │              │
└─────────┘         └──────────────┘              ▼
                                        ┌─────────────────────────────┐
                                        │  Generate Query Embedding   │
    ▼                        ▼          │  (OpenAI Embeddings API)    │
                                        └─────────────────────────────┘
┌─────────┐         ┌──────────────┐              │
│clarifi- │         │ guardrail    │              │
│cation   │         │              │              ▼
│         │         │Response:     │    ┌─────────────────────────────┐
```

```
│Response:│         │"I can't      │    │  HYBRID SEARCH (PostgreSQL) │
│"Could   │         │share that"   │    │  • Semantic (cosine sim)    │
│you      │         └──────────────┘    │  • Keyword (tsvector)       │
│clarify?"│                              │  • Filter by company_id     │
└─────────┘                ▼             │  • Threshold: 0.5           │
                    ┌──────────────┐    │  • Limit: 3 results         │
    ▼               │ human_support│    └─────────┬───────────────────┘
┌─────────┐         │              │              │
│ALL      │         │Response:     │    ┌─────────┴─────────┐
│INTENTS  │         │"Please       │    │                   │
│         │         │provide email"│    ▼                   ▼
│Save to  │         └──────────────┘ ┌──────┐          ┌──────────┐
│database │                          │FOUND │          │NOT FOUND │
│& return │                ▼          │      │          │          │
│to user  │         ┌──────────────┐ └──┬───┘          └────┬─────┘
└─────────┘         │human_support_│    │                   │
                    │email         │    │                   │
                    │              │    ▼                   ▼
                    │Response:     │ ┌─────────┐      ┌──────────┐
                    │"Thank you!   │ │RESULTS  │      │FALLBACK  │
                    │Agent will    │ │CONTEXT  │      │RESPONSE  │
                    │contact you"  │ └────┬────┘      └────┬─────┘
                    └──────────────┘      │                │
                                          ▼                ▼
                                ┌──────────────────────────────┐
                                │  AI RESPONSE GENERATION      │
                                │  (OpenAI Chat Completion)    │
                                │                              │
                                │  Model: AI_MODEL             │
                                │  Temperature: AI_TEMPERATURE │
                                │  Max Tokens: AI_MAX_TOKENS   │
                                │                              │
                                │  System Prompt includes:     │
                                │  • Platform description      │
                                │  • Core capabilities         │
                                │  • Security guardrails       │
                                │  • Knowledge base context    │
                                │  • Custom instructions       │
                                └──────────────┬───────────────┘
                                               │
                                               ▼
                                ┌──────────────────────────────┐
                                │  SAVE & RETURN RESPONSE      │
                                │  • Save assistant message    │
                                │  • Update conversation       │
                                │  • Return to user            │
                                └──────────────────────────────┘
```

Lightweight Flow Diagram

```
USER QUERY
    │
    ▼
LOAD CONTEXT (History + Company)
    │
    ▼
🧭 INTENT CLASSIFIER (OpenAI)
    │
    ├─────────────────────┬──────────────────┐
    │                     │                  │
    ▼                     ▼                  ▼
SIMPLE                KNOWLEDGE          GUARDRAIL
(greeting,            (RAG Flow)         (Security)
personality,              │                  │
clarification,            ▼                  │
support)          Generate Embedding         │
    │                     │                  │
    │                     ▼                  │
    │             Hybrid Search              │
    │             (Semantic + Keyword)       │
    │                     │                  │
    │             ┌───────┴────────┐         │
    │             ▼                ▼         │
    │          Found            Not Found    │
    │             │                │         │
    │             ▼                ▼         │
    │       Use Results      Fallback        │
    │             │           Message        │
    │             └────┬─────────┘           │
    │                  ▼                     │
    └──────────► AI RESPONSE ◄──────────────┘
                 (OpenAI)
                     │
                     ▼
              SAVE & RETURN
```

Key Components

1. Intent Classification

Purpose: Optimize resource usage by identifying query type before expensive RAG operations

Intents:

- `knowledge` : Questions requiring knowledge base lookup (triggers RAG)
- `greeting` : Simple greetings (direct response: "Hello!")
- `personality` : Questions about assistant identity (direct response with name)
- `clarification` : Unclear/incomplete queries (ask for clarification)
- `guardrail` : Requests for sensitive information (security refusal)
- `human_support_request` : User wants human agent (ask for email)
- `human_support_email` : User provides email (confirm handoff)

Benefits:

- Reduces OpenAI API calls for simple queries
- Prevents unnecessary database searches
- Improves response time for common interactions

2. RAG (Retrieval-Augmented Generation) Pipeline

Triggered by: `knowledge` intent only

Steps:

1. **Generate Embedding:** Convert query to vector (OpenAI Embeddings API)
2. **Hybrid Search:**
    - Semantic: Cosine similarity between embeddings
    - Keyword: PostgreSQL full-text search (tsvector)
    - Company Filter: Only search within user's organization
    - Threshold: 0.5 (similarity score)
    - Limit: Top 3 results
3. **Context Assembly:** Build system prompt with retrieved knowledge
4. **AI Generation:** OpenAI generates response using context

3. Security Guardrails

Implemented at two levels:

Intent Classifier:

- Detects requests for API keys, passwords, tokens, env vars
- Returns `guardrail` intent

System Prompt:

- Explicit instructions to never expose secrets
- Safe architectural guidance allowed
- Redaction of sensitive configuration

Refusal Message:

> *"I can help with documentation or implementation guidance, but I can't share credentials or confidential configuration. Please contact your system administrator or support for access."*

4. Configuration (Environment Variables)

```
AI_MODEL=gpt-4o-mini              # Model for all OpenAI calls
AI_TEMPERATURE=0.7                # Response creativity (0-1)
AI_MAX_TOKENS=1000                # Maximum response length
CHAT_HISTORY_LENGTH=2             # Messages for context
ORGANIZATION_NAME=Your Organization
ASSISTANT_NAME=AI Assistant
PLATFORM_DESCRIPTION=...          # Custom platform description
```

5. Chat History Context

- Retrieves last N messages (default: 2)
- Provides conversation continuity

- Used by both intent classifier and response generator
- Enables multi-turn interactions (e.g., email collection)

Flow Statistics

API Calls per Query Type

Query Type	Intent Classification	Embedding Generation	AI Response	Total
Greeting	✓	-	-	1
Personality	✓	-	-	1
Clarification	✓	-	-	1
Guardrail	✓	-	-	1
Support Request	✓	-	-	1
Knowledge (found)	✓	✓	✓	3
Knowledge (not found)	✓	✓	✓	3

Performance Optimization

- **Before Intent Classifier:** Every query → 3 API calls (embed + search + generate)
- **After Intent Classifier:** Simple queries → 1 API call (60-80% reduction for typical usage)

Database Schema (Relevant Tables)

`vezlo_knowledge`

- `id` (integer): Primary key
- `uuid` (uuid): External identifier
- `company_id` (integer): Organization filter
- `title` , `description` , `content` : Knowledge content
- `embedding` (vector): Semantic search vector
- `metadata` (jsonb): Additional data

`vezlo_conversations`

- `id` (integer): Primary key
- `uuid` (uuid): External identifier
- `company_id` (integer): Organization filter
- `creator_id` (integer): User reference
- `title` : Conversation name

`vezlo_messages`

- `id` (integer): Primary key
- `uuid` (uuid): External identifier
- `conversation_id` (integer): Parent conversation
- `sender_type` : 'user' | 'assistant'
- `content` : Message text

- `parent_message_id`: Threading support

Error Handling

No Knowledge Found

When RAG search returns 0 results:

> *"I'm sorry, I couldn't find the requested information in my knowledge base. Please contact support for further assistance or check if the information might be available in other resources."*

LLM Behavior

- System prompt instructs LLM to ONLY use knowledge base context
- If no context provided, LLM must use fallback message
- Prevents hallucination from general training knowledge

Future Improvements

1. Streaming Responses (Planned)

- Server-Sent Events (SSE)
- Real-time token streaming
- Better UX for long responses

2. Knowledge Chunks Architecture (Planned)

- New `knowledge_chunks` table
- Chunk-level embeddings
- More precise source citations
- Better relevance scoring

3. Manual Tool Calling (Discussed)

- Intent dispatcher to server-side functions
- Extensible for future capabilities (scheduling, ticketing, etc.)
- Full control over execution flow

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Phase: 3 (Intent Classification + RAG + Guardrails)